

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO1-AT3 - CODING CHALLENGE

View Only

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QUESTIONS

Q1

Smart University Payroll
Management System
25 marks

Q2

Smart Vehicle Insurance Claim
System
25 marks

2/2 answered

Q1

Smart University Payroll Management
System

25

marks

A university employs three categories of staff: Teaching Faculty, Technical Staff, and Visiting Faculty. The payroll system should calculate the monthly salary based on the employee type while maintaining a common employee hierarchy. Requirements Create a superclass Employee with the following private data members: employeeId employeeName department basicSalary Provide: Default constructor Parameterized constructor Getter and Setter methods displayEmployee() calculateSalary() Create the following subclasses: TeachingFaculty teachingHours incentivePerHour Salary: basicSalary + (teachingHours × incentivePerHour) TechnicalStaff overtimeHours overtimeRate Salary: basicSalary + (overtimeHours × overtimeRate) VisitingFaculty lecturesHandled paymentPerLecture Salary: lecturesHandled × paymentPerLecture Override calculateSalary() in all subclasses. Create an array of Employee objects and demonstrate runtime polymorphism. Display: Employee details Monthly salary Highest-paid employee Average salary Search an employee using the employee ID.

Test Case 1 (Normal Case)

Input

3

TeachingFaculty

101

Dr. Arun

CSE

50000

20

500

TechnicalStaff

102

Mr. Ravi

ECE

35000

15

300

VisitingFaculty

103

Dr. Kumar

IT

0

40

1200

Search Employee ID:

102

Expected Output

Employee Details

Employee ID : 101

Name : Dr. Arun

Department : CSE

Monthly Salary : 60000.0

Employee ID : 102

Name : Mr. Ravi

Department : ECE

Monthly Salary : 39500.0

Employee ID : 103

Name : Dr. Kumar

Department : IT

Monthly Salary : 48000.0

Highest Paid Employee

Employee ID : 101

Salary : 60000.0

Average Salary

49166.67

Employee Found

Employee ID : 102

Name : Mr. Ravi

Salary : 39500.0

Your Code

```
1 import java.util.*;
2 class Employee{
3     private int id;
4     private String name, dept;
5     protected double b;
6     Employee() {}
7     Employee(int i,String n,String d,douk
8         id=i;name=n;dept=d;b=x;
9     }
10    public double calculateSalary(){
11        return b;
12    }
13    public void displayEmployee(){
14        System.out.println("\nEmployee ID
15        System.out.println("Name : "+name
16        System.out.println("Department :
17        System.out.println("Monthly Salar
18        System.out.println();
19    }
20    public static void main(String[] args
21        Scanner s = new Scanner(System.in
22        int n =s.nextInt();Employee[] a=
23        for(int i=0;i<n;i++){
24            String t=s.next();int id=s.ne
25            s.nextLine();
26            String name=s.nextLine().trim
27            String d=s.next();
28            double b=s.nextDouble();
29            int x=s.nextInt();
30            double y=s.nextDouble();
```

```

31
32         if(t.equals("TeachingFaculty")
33             a[i]=new TeachingFaculty(
34         else if(t.equals("TechnicalSt
35             a[i]=new TechnicalStaff(i
36         else{
37             a[i]=new VisitingFaculty(
38         }}
39     System.out.println("Employee Deta
40     double sum=0;Employee hi=a[0];
41
42     for(Employee e:a){
43         e.displayEmployee();
44         sum+=e.calculateSalary();
45         if(e.calculateSalary()>hi.cal
46             hi=e;
47     }
48     System.out.println();
49     System.out.println("Highest Paid
50     System.out.println("Employee ID :
51     System.out.println("Salary : "+hi
52     System.out.println();
53     System.out.println("Average Salar
54     System.out.printf("%.2f\n", sum/n)
55
56
57     int search=s.nextInt();
58     boolean found = false;
59
60     for(Employee e:a){
61         if(e.id==search){
62             System.out.println();
63             System.out.println("Emplc
64             System.out.println("Emplc
65             System.out.println("Name
66             System.out.println("Salar
67             found=true;
68             break;
69         }
70     }
71     if(!found)
72         System.out.println("Employee Not
73     s.close();
74 }}
75 class TeachingFaculty extends Employee{
76     double h,r;
77     TeachingFaculty(int i,String n,String
78         super(i,n,d,b);

```

```
79         h=x;
80         r=y;
81     }
82     public double calculateSalary() {
83         return b+h*r;
84     }
85 }
86 class TechnicalStaff extends Employee{
87     double h,r;
88     TechnicalStaff(int i,String n,String
89         super(i,n,d,b);
90         h=x;
91         r=y;
92     }
93     public double calculateSalary() {
94         return b+h*r;
95     }
96 }
97 class VisitingFaculty extends Employee{
98     int l;
99     double p;
100     VisitingFaculty(int i,String n,String
101         super(i,n,d,b);
102         l=x;
103         p=y;
104     }
105     public double calculateSalary() {
106         return l*p;
107     }
108 }
```



Input

3

TeachingFaculty

101



Output

Employee Details

Employee ID : 101

Name : Dr.Arun

Department : CSE

Monthly Salary : 60000.0



Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



0.00 / 25 marks

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